# Exercises

June 12, 2024

## 1 Basic probability concepts

1.1 Given $P(A) = 0.4, P(B) = 0.5$. If $A$ and $B$ are independent, find $P(AB)$.

1.2 Given $P(A) = 0.4, P(B) = 0.5$, and $P(AB) = 0.05$.

a. Find $P(A|B)$.

b. In this problem, are $A$ and $B$ independent?

1.3 Which pairs of events are independent?

a. $P(A) = 0.6, P(B) = 0.4, P(AB) = 0.24$.

b. $P(A) = 0.9, P(B) = 0.2, P(AB) = .18$.

c. $P(A) = 0.5, P(B) = 0.7, P(AB) = 0.25$.

1.4 The probability that a student has a Visa card (event V) is 0.73. The probability that a student has a MasterCard (event M) is 0.18. The probability that a student has both cards is 0.03.

a. Find the probability that a student has either a Visa card or a MasterCard.

b. In this problem, are V and M independent?

1.5 Ninety percent of flights depart on time. Eighty percent of flights arrive on time. Seventy-five percent of flights depart on time and arrive on time.

a. You are meeting a flight that departed on time. What is the probability that it will arrive on time?

b. You have met a flight, and it arrived on time. What is the probability that it departed on time ?

c. Are the events, departing on time and arriving on time, independent ?

1.6 Over 1,000 people try to climb Mt. Everest every year. Of those who try to climb Everest, 31 percent succeed. The probability that a climber is at least 60 years old is 0.04. The probability that a climber is at least 60 years old and succeeds in climbing Everest is 0.005.

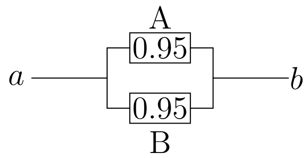
a. Find the probability of success, given that a climber is at least 60 years old.

b. Is success in climbing Everest independent of age? (See The New York Times, August 21, 2007, p. D3).

1.7 50 percent of the customers at Pizza Palooza order a square pizza, 80 percent order a soft drink, and 40 percent order both a square pizza and a soft drink. Is ordering a soft drink independent of ordering a square pizza? Explain.

1.8 Automobiles are equipped with redundant braking circuits; their brakes fail only when all circuits fail. Consider one with two redundant braking circuits, each having a reliability of 0.95. Determine the system reliability assuming that these circuits act independently.

Comment: Systems of this type are graphically represented as in below figure, in which the circuits (A and B) have a parallel arrangement. The path to success is broken only when breakdowns of both A and B occur. Consider the path $a \to b$ as the 'path to success'.




1.9 On a stretch of highway, the probability of an accident due to human error in any given minute is $10^{-5}$, and the probability of an accident due to mechanical breakdown in any given minute is $10^{-7}$. Assuming that these two causes are independent:

a. Find the probability of the occurrence of an accident on this stretch of highway during any minute.

b. In this case, can the above answer be approximated by P(accident due to human error) P(accident due to mechanical failure)? Explain.

c. If the events in succeeding minutes are mutually independent, what is the probability that there will be no accident at this location in a year?

1.10 Rapid transit trains arrive at a given station every five minutes and depart after stopping at the station for one minute to drop off and pick up passengers. Assuming trains arrive every hour on the hour, what is the probability that a passenger will be able to board a train immediately if he or she arrives at the station at a random instant between 7:54 a.m. and 8:06 a.m.?

1.11 A machine part may be selected from any of three manufacturers with probabilities $p_1 = 0.25, p_2 = 0.5$ and $p_3 = 0.25$. The probabilities that it will function properly during a specified period of time are $0.2, 0.3,$ and $0.4$, respectively, for the three manufacturers. Determine the probability that a randomly chosen machine part will function properly for the specified time period.

1.12 Consider the possible failure of a transportation system to meet demand during rush hour.

a. Determine the probability that the system will fail if the probabilities shown in Table are known.

| Demand level | $P$(level) | $P$(system failure — level) |
|---|---|---|
| Low | 0.6 | 0 |
| Medium | 0.3 | 0.1 |
| High | 0.1 | 0.5 |

b. If system failure was observed, find the probability that a 'medium' demand level was its cause.

1.13 A drug test for athletes has a 5 percent false positive rate and a 10 percent false negative rate. Of the athletes tested, 4 percent have actually been using the prohibited drug. If an athlete tests positive, what is the probability that the athlete has actually been using the prohibited drug?

1.14 Half of a set of the parts are manufactured by machine A and half by machine B. Four percent of all the parts are defective. Six percent of the parts manufactured on machine A are defective. Find the probability that a part was manufactured on machine A, given that the part is defective.

1.15 An airport gamma ray luggage scanner coupled with a neural net artificial intelligence program can detect a weapon in suitcases with a false positive rate of 2 percent and a false negative rate of 2 percent. Assume a .001 probability that a suitcase contains a weapon. If a suitcase triggers the alarm, what is the probability that the suitcase contains a weapon?

# 2 Random variables and probability distribution

2.1 Find $k$ so that the function given by

$$p(x) = \frac{k}{x+1}, \quad x = 1, 2, 3, 4$$

is a probability mass function. Find the cumulative distribution function $F(x)$.

2.2 A random variable $X$ has the following probability mass function:

| $x$ | -5 | 0 | 3 | 6 |
|---|---|---|---|---|
| $P(X = x)$ | 0.2 | 0.1 | 0.4 | 0.3 |

Find the cumulative distribution function $F(x)$.

2.3 The cumulative distribution function $F(x)$ of a random variable $X$ is given by

$$F(x) = \begin{cases} 0, & x < -1 \\ 0.2, & -1 \leq x < 3 \\ 0.8, & 3 \leq x < 9 \\ 1, & x \geq 9. \end{cases}$$

Write down the values of the random variable $X$ and the corresponding probabilities, $p(x)$.

2.4 Let the function

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the value of $c$ so that $f(x)$ is a density function.

b. Compute $P(2 < X < 3)$.

c. Find the distribution function $F(x)$.

2.5 The amount of time, in hours, that a machine functions before breakdown is a continuous random variable with pdf

$$f(x) = \begin{cases} \dfrac{1}{120} e^{-x/120}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

a. What is the probability that this machine will function between 98 and 145 hours before breaking down?

b. What is the probability that it will function less than 160 hours?

2.6 Let $T$ denote the life (in months) of a light bulb and let

$$f(t) = \begin{cases} \dfrac{1}{15} - \dfrac{t}{450}, & 0 \leq t \geq 30 \\ 0, & \text{elsewhere.} \end{cases}$$

a. Determine the probability that the light bulb will last at least 15 months.

b. A light bulb has already lasted 15 months. What is the probability that it will survive another month?

2.7 From a bag of fruits which contain 4 oranges, 2 apples, and 2 bananas, a random sample of 4 fruits are selected. Let $X$ be the number of oranges and $Y$ be the number of apples.

a. Find the joint distribution of $X$ and $Y$.

b. Find the marginal PMFs of $X$ and $Y$.

c. Find whether $X$ and $Y$ are independent random variables.

2.8 Suppose that $X$ and $Y$ have the following joint probability distribution

| $X$ \ $Y$ | -2 | 0 | 1 | 4 |
|---|---|---|---|---|
| 1 | 0.3 | 0.1 | 0.0 | 0.2 |
| 3 | 0.0 | 0.2 | 0.1 | 0.0 |
| 5 | 0.1 | 0.0 | 0.0 | 0.0 |

a. Find the marginal pmf of the random variables $X$ and $Y$.

b. Find $P(X > 1 | Y > 0)$.

2.9 The joint probability distribution of a pair of random variables is given by the following table:

| $Y$ \ $X$ | 1 | 2 | 3 |
|---|---|---|---|
| 1 | 0.25 | 0.05 | 0.25 |
| 2 | 0.35 | 0.05 | 0.05 |

a. Find the marginal density functions of $X$ and $Y$.

b. Find the conditional distribution of $X$ given $Y = 1$.

c. Find $P(X + Y < 4)$.



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2.10 A person sends 1, 2, or 3 text messages an hour, and when she sends it, a text message has 40, 80, or 120 characters. The following table provides the corresponding probabilities for all nine possible cases:

| | 1 message | 2 messages | 3 messages |
|---|---|---|---|
| 40 Characters | 0.05 | 0.10 | 0.05 |
| 80 Characters | 0.10 | 0.15 | 0.15 |
| 120 Characters | 0.15 | 0.20 | 0.05 |

Let the random variable $X$ represent the number of messages sent and the random variable $Y$ represent the length of a message, measured in characters.

a. Find the marginal pmf of the random variables $X$ and $Y$.

b. Find $P(XY > 200)$.

2.11 Let $X$ and $Y$ be two jointly continuous random variables with joint PDF

$$f(x,y) = \begin{cases} c(1-x)(1-y), & -1 \le x \le 1, -1 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.

b. Find the marginal PDF $f_X(x)$.

c. Find $P\left(0 \le X \le 1, 0 \le Y \le 1\right)$.

d. Find $P(X < Y)$.

2.12 Let $X$ and $Y$ be two jointly continuous random variables with joint PDF

$$f(x,y) = \begin{cases} cxy, & 0 \le x \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.

b. Find the marginal PDF $f_X(x)$.

c. Find $P\left(1.5 \le X \le 1.75, Y > 1\right)$.

2.13 Let $X$ and $Y$ be two jointly continuous random variables with joint PDF

$$f(x,y) = \begin{cases} c(4x+2y+1), & 0 \le x \le 2, 0 \le y \le 2 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.

b. Find the marginal PDF $f_X(x)$.

c. Find $f(y|x)$.

2.14 Suppose the joint pdf of the random variables $X$ and $Y$ is as follows:

$$f(x,y) = \begin{cases} c(1-x), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.

b. Find the marginal PDF $f_X(x)$.

c. Find $P(X \leq 0.5, 0.4 \leq Y \leq 0.7)$.

2.15 The joint pdf of random variables $X, Y$ is as follows:

$$f(x,y) = \begin{cases} c(x+y), & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

a. Find the constant $c$.

b. Find the marginal PDF $f_X(x)$.

c. Find the conditional density function $f(x|y)$

2.16 The joint pdf of random variables $X, Y$ is as follows:

$$f(x,y) = \begin{cases} y^2 e^{-y(x+1)}, & 0 \leq x, 0 \leq y \\ 0, & \text{otherwise} \end{cases}$$

Find the marginal PDFs $f_X(x), f_Y(y)$.

2.17 The joint pdf of random variables $X, Y$ is as follows:

$$f(x,y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq x \leq y \\ 0, & \text{otherwise} \end{cases}$$

Find the marginal PDFs $f_X(x), f_Y(y)$.

# 3 Expectations and Moments

3.1 Consider rolling an unfair (biased) die, with the following probabilities for its possible outcomes:

$$P(1) = P(2) = P(3) = 0.1, P(4) = P(5) = 0.2, P(6) = 0.3$$

Determine its mean and variance.

3.2 The random variable $X$ representing the number of errors per 100 lines of software code has the following probability distribution:

| $x$ | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| $p(x)$ | 0.01 | 0.25 | 0.4 | 0.3 | 0.04 |

Determine mean and variance of $X$.

3.3 The probability distribution of a discrete random variable $X$ is given in the following table:

| $x$ | -1 | 0 | 2 | 5 | 6 |
|---|---|---|---|---|---|
| $p(x)$ | 0.1 | 0.05 | 0.25 | 0.4 | 0.2 |

Find $E(X)$ and $\text{var}(X)$.

3.4 A panel of meteorological and civil engineers studying emergency evacuation plans for Florida's Gulf Coast in the event of a hurricane has estimated that it would take between 13 and 18 hours to evacuate people living in low-lying land with the probabilities shown in Table

| Time to evacuate (nearest hour) | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|
| Probability | 0.04 | 0.25 | 0.4 | 0.18 | 0.1 | 0.03 |

Calculate the mean and standard deviation of the probability distribution of the evacuation times.

3.4 For a laboratory assignment, if the equipment is working, the density function of the observed outcome $X$ is

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the variance and standard deviation of $X$.

3.5 The weekly demand for a certain drink, in thousands of liters, at a chain of convenience stores is a continuous random variable $g(X) = X^2 + X - 2$, where $X$ has the density function

$$f(x) = \begin{cases} 2(x-1), & 1 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

3.6 In a course, the number of students who go to the office hours to ask questions during a semester is a random variable, whose mean and standard deviation are 18 and 2.5, respectively. Determine the probability that there will be more than 8, but fewer than 28 students going to the office hours.

3.7 To find out the prevalence of smallpox vaccine use, a researcher inquired into the number of times a randomly selected 200 people aged 16 and over in an African village had been vaccinated. He obtained the following figures: never, 17 people; once, 30; twice, 58; three times, 51; four times, 38; five times, 7. Assuming these proportions continue to hold exhaustively for the population of that village, what is the expected number of times those people in the village had been vaccinated, and what is the standard deviation?

3.8 Assume the length $X$, in minutes, of a particular type of telephone conversation is a random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{3}e^{-x/3}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

a. Determine the mean length $E(X)$ of this type of telephone conversation.

b. Find the variance and standard deviation of $X$.

c. Find $E[(X+5)^2]$.

3.9 Let $X$ be a continuous random variable with joint PDF

$$f(x) = \begin{cases} \dfrac{5}{x^6}, & x \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

a. What bound does Chebyshev's inequality give for the probability $P(X \geq 2.5)$?

b. For what value of $a$ can we say $P(X \geq a) \leq 15\%$?

3.10 A convenience store has two separate locations where customers can be checked out as they leave. These locations each have two cash registers and two employees who check out customers. Let $X$ be the number of cash registers being used at a particular time for location 1 and $Y$ the number being used at the same time for location 2. The joint probability function is given by

| $X \backslash Y$ | 0 | 1 | 2 |
|---|---|---|---|
| 0 | 0.12 | 0.04 | 0.04 |
| 1 | 0.08 | 0.19 | 0.05 |
| 2 | 0.06 | 0.12 | 0.3 |

a. Find the marginal density of both $X$ and $Y$ as well as the probability distribution of $X$ given $Y = 2$.

b. Find $E(X)$ and $\text{var}(X)$.

c. Find $E(X|Y = 2)$.

# 4 Functions of random variables

# 5 Useful probability distributions

5.1 The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that

a. at least 10 survive,

b. from 3 to 8 survive,

c. exactly 5 survive?

5.2 A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 3%.

a. The inspector randomly picks 20 items from a shipment. What is the probability that there will be at least one defective item among these 20?

b. Suppose that the retailer receives 10 shipments in a month and the inspector randomly tests 20 devices per shipment. What is the probability that there will be exactly 3 shipments each containing at least one defective device among the 20 that are selected and tested from the shipment?

5.3 In a certain city district, the need for money to buy drugs is stated as the reason for 75% of all thefts. Find the probability that among the next 5 theft cases reported in this district,

a. exactly 2 resulted from the need for money to buy drugs;

b. at most 3 resulted from the need for money to buy drugs.

5.4 According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 8 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is

a. anywhere from 2 to 5;

b. fewer than 3.

5.5 One prominent physician claims that 70% of those with lung cancer are chain smokers. If his assertion is correct,

a. find the probability that of 10 such patients recently admitted to a hospital, fewer than half are chain smokers;

b. find the probability that of 20 such patients recently admitted to a hospital, fewer than half are chain smokers.

5.6 It is known that 60% of mice inoculated with a serum are protected from a certain disease. If 5 mice are inoculated, find the probability that

a. none contracts the disease;

b. fewer than 2 contract the disease;

c. more than 3 contract the disease.

5.7 A foreign student club lists as its members 2 Canadians, 3 Japanese, 5 Italians, and 2 Germans. If a committee of 4 is selected at random, find the probability that

a. all nationalities are represented;

b. all nationalities except Italian are represented.

5.8 A case of wine has 12 bottles, 3 of which contain spoiled wine. A sample of 4 bottles is randomly selected from the case.

a. Find the probability distribution for $X$, the number of bottles of spoiled wine in the sample.

b. What are the mean and variance of $X$?

5.9 A company has five applicants for two positions: two women and three men. Suppose that the five applicants are equally qualified and that no preference is given for choosing either gender. Let $X$ equal the number of women chosen to fill the two positions.

a. Write the formula for $p(X = x)$, the probability distribution of $x$.

b. What are the mean and variance of this distribution?

5.10 Consider a plant manufacturing IC chips of which 5% are expected to be defective. The chips are packed 30 to a box. A sample of size 10 is drawn without replacement from each box. If more than one defective chip is found in the sample, the box is rejected and subjected to a complete inspection. What is the probability that a box will be rejected?

5.11 Components are packed in boxes and a simple random sample is chosen from each box. The selected components are inspected to decide whether to accept or reject the box. If more than 10% of the chips in the sample are defective, the box is rejected. In each of the following circumstances, estimate the probability

of this happening when a box contains 5% defectives:

a. A sample of 10 is selected from a box of 40

b. A sample of 10 is selected from a box of 5000

5.12 The manufacturer of disk drives in one of the well-known brands of microcomputers expects 1% of the disk drives to malfunction during its warranty period. Calculate the probability that in a sample of 100 disk drives, not more than three will malfunction.

5.13 The average rate of job submissions in a busy computer center is 4 per minute. If it can be assumed that the number of submissions per minute interval is Poisson distributed, calculate the probability that

a. at least two jobs will be submitted in any minute,

b. no job will be submitted in any minute,

c. not more than one job will be submitted in any one-minute interval.

5.14 A new user of a mobile telephone receives an average of three messages per day. If it can be assumed that the arrival pattern of these messages is Poisson, calculate the following probabilities:

a. Exactly three messages will be received in any day.

b. More than three messages will arrive in a day.

c. A day will pass without a message.

d. If 1 day has passed without a message, six messages will be received the next day.

5.15 The number of telephone calls that arrive at an exchange is often modeled as a Poisson random variable. Assume that on an average there are 10 calls per hour. What is the probability that

a. there are exactly five calls in 1 h?

b. there are three or fewer calls in 1 h?

5.16 When a computer disk manufacturer tests a disk, they write to the disk and then test it using a certifier. The certifier counts the number of missing pulses or errors. It can be assumed that the number of errors on a test area on a disk has a Poisson distribution with $\lambda = 0.2$.

a. What is the expected number of errors per test area?

b. What percentage of test areas have two or fewer errors?

5.17 Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

a. less than 5 minutes for a bus;

b. at least 12 minutes for a bus.

5.18 Suppose $X$ follows a continuous uniform distribution from 0 to 5. Determine the conditional probability, $P(X < 3.5|X \geq 1)$.

5.19 An automatic machine fills distilled water in 500-ml bottles. Actual volumes are normally distributed about a mean of 500 ml and their standard deviation is 20 ml.

a. What proportion of the bottles are filled with water outside the tolerance limit of 475 ml to 525 ml?

b. To what value does the standard deviation need to be increased if 99% of the bottles must be within tolerance limits?

5.20 Intelligence quotients (IQs) measured on the Stanford Revision of the Binet–Simon Intelligence Scale are normally distributed with a mean of 100 and a standard deviation of 16. Determine the percentage of people who have IQs between 115 and 140.

5.20 The time needed to complete a final examination in a particular college course is normally distributed with a mean of 80 minutes and a standard deviation of 10 minutes. Answer the following questions.

a. What is the probability of completing the exam in one hour or less?

b. What is the probability that a student will complete the exam in more than 60 minutes but less than 75 minutes?

c. Assume that the class has 60 students and that the examination period is 90 minutes in length. How many students do you expect will be unable to complete the exam in the allotted time?

5.21 The average stock price for companies making up the S&P 500 is $30, and the standard deviation is $8.20 (BusinessWeek, Special Annual Issue, Spring 2003). Assume the stock prices are normally distributed.

a. What is the probability a company will have a stock price of at least $40?

b. What is the probability a company will have a stock price no higher than $20?

c. How high does a stock price have to be to put a company in the top 10%?

5.22 The average length of steel nails is 5 centimeters, with a standard deviation of 0.05 centimeters. Assuming that the lengths are normally distributed, what percentage of the nails are

a. longer than 5.05 centimeters?

b. between 4.95 and 5.05 centimeters in length?

c. shorter than 4.90 centimeters?

5.23 A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?

5.24 If 20% of the residents in a city prefer a white cell phone over any other color available, what is the probability that among the next 1000 cell phone purchased in that city

a. between 170 and 185 inclusive will be white?

b. at least 210 but not more than 225 will be white?

5.25 The systolic blood pressure X of 30-year-old men is approximately normally distributed with a mean of 123 mmHg and standard deviation of 6 mmHg.

a. Find the probability that the blood pressure of a randomly selected 30-year-old man exceeds 127 mmHg.

b. Out of 250 randomly selected men of 30 years old, find the probability that the blood pressure of at least 80 of them is more than 127.

Use the normal approximation to the binomial to find the answer.

5.26 The reliability of an electrical fuse is the probability that a fuse, chosen at random from production, will function under its designed conditions. A random sample of 1000 fuses was tested and 27 defectives were observed. Calculate the approximate probability of observing 27 or more defectives, assuming that the fuse reliability is 0.98.

5.27 A Myrtle Beach resort hotel has 120 rooms. In the spring months, hotel room occupancy is approximately 75%.

a. What is the probability that at least half of the rooms are occupied on a given day?

b. What is the probability that 100 or more rooms are occupied on a given day?

c. What is the probability that 80 or fewer rooms are occupied on a given day?

5.28 According to the American Red Cross, 7% of people in the United States have blood type O-negative. What is the probability that, in a simple random sample of 500 people in the United States, fewer than 30 have blood type O-negative?

5.29 According to American Airlines, Flight 215 from Orlando to Los Angeles is on time 90% of the time. Randomly select 150 flights and use the normal approximation to the binomial to

a. approximate the probability that exactly 130 flights are on time.

b. approximate the probability that at least 130 flights are on time.

c. approximate the probability that fewer than 125 flights are on time.

d. approximate the probability that between 125 and 135 flights, inclusive, are on time.

# 6 Limit theorems

6.1 The times spent by customers coming to a certain gas station to fill up can be viewed as independent random variables with a mean of 3 min and a variance of 1.5 min. Approximate the probability that 75 customers in this gas station will spend a total time of less than 3 hours.

6.2 The local farm packs its tomatoes in crates. Individual tomatoes have mean weight of 10 ounces and standard deviation 3 ounces. Find the probability that a crate of 50 tomatoes weighs between 480 and 510 ounces.

6.3 The waiting time on the cashier's line at the school cafeteria is exponentially distributed with mean 2 minutes. Use the central limit theorem to find the approximate probability that the average waiting time is more than 2.5 minutes for a group of 20 people. Use R and compare with the exact probability.

6.4 If the distribution of scores of all students in an examination has a mean of 296 and a standard deviation of 14, what is the probability that the combined gross score of 49 randomly selected students is less than 14,250?

6.5 An assembly worker has to assemble 50 parts. The time to assemble each is independent and identically distributed with mean 8 min, and standard deviation 2 min. Approximate the probability that all 50 parts will be assembled in less than 7 h. (0.9207)

6.6 The number of the typographical errors on a book page is assumed to be independent and Poisson distributed with mean 2 for each page. If we select 40 pages from this book randomly, find the probability that there are at most 100 errors. Find this probability also by approximation. (0.9875)

6.7 Car mufflers are constructed by nearly automatic machines. One manufacturer finds that, for any type of car muffler, the time for a person to set up and complete a production run has a normal distribution with mean 1.82 hours and standard deviation 1.20. What is the probability that the mean of the next 40 runs will be from 1.65 to 2.04 hours. (0.6917)

6.8 An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that 16 bulbs will have an average life of less than 775 hours. (0.0062)

6.9 Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute. (0.0008)

6.10 The length (cm) of a manufactured machine part is a random variable $X$ with the pdf

$$f(x) = \begin{cases} 3(1-x)^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability that 16 parts will have an average length from 0.125 to 0.375. (0.9876)